

TrES-1b

R-1862

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_150716 · created 2026-08-03T09:41:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457219.8316 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.052
Transit depth (Rp/Rs)^2	1.26 +/- 1.18 [%]
Orbital Inclination (inc)	88.66 +/- 1.44
Airmass coefficient 1 (a1)	0.964 +/- 0.02
Airmass coefficient 2 (a2)	0.03 +/- 0.18
Scatter in the residuals of the lightcurve fit is	2.1 %
Best Comparison Star	#2 - [495, 206]
Optimal Aperture	2.61
Optimal Annulus	12.38
Transit Duration (day)	0.096 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.052 vs 0.1358 (deviation 0.46σ)
Duration fit vs published	0.096 d vs 0.1042 d (deviation 0.82σ)
Mid-time O–C	1.8 min
Depth SNR	2.2
Residual scatter	2.1 %

Light curve

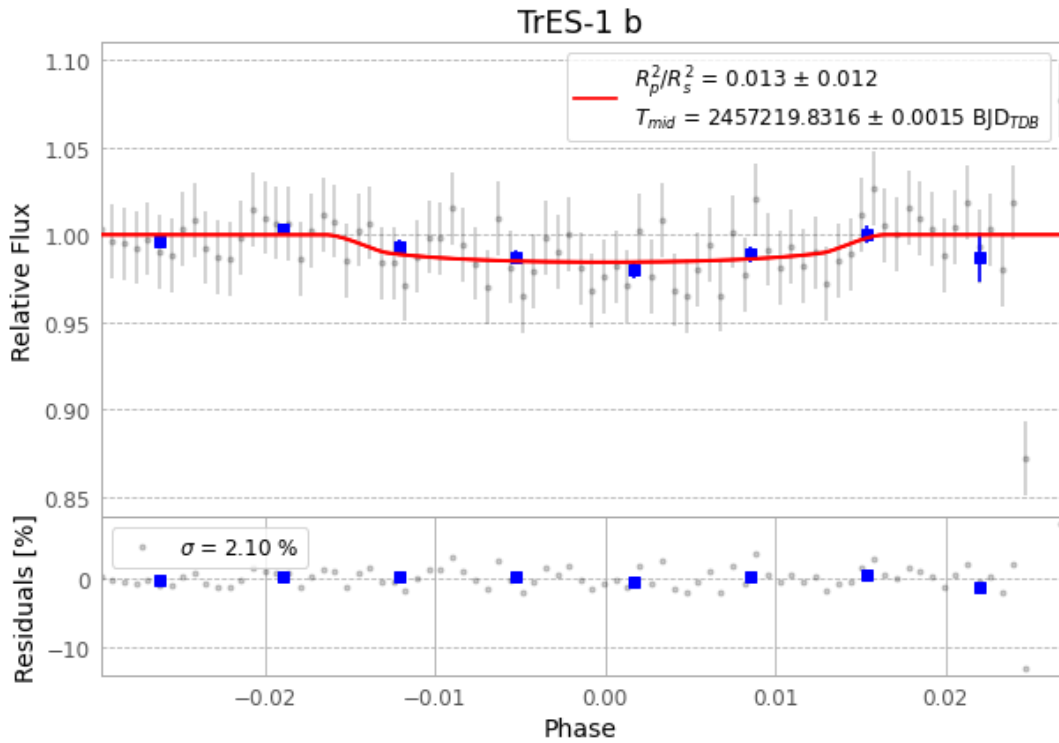


Figure 1 — FinalLightCurve_TrES-1 b_2015-07-15.png (SHA-256 e13ea5504484a8e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [205, 229], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0764e0a4afcaeead...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1150716054212.FITS → TRES-1150716101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-150716.log (45b844968394...), FinalLightCurve_TrES-1 b_2015-07-15.png (e13ea5504484...), FinalLightCurve_TrES-1 b_2015-07-15.csv (aee51b95072d...)

Compute resources

Wall time 460.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 09:41Z.